Debug With Shubham Notes

TCS NQT Test Curriculum 2025

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TCS NQT Foundation Aptitude Curriculum

Number System

Mensuration

Arithmetic Ability

Elementary Statistics

Data Interpretation

Simplification & Approximation

Number System

1- Divisiblity

2- Numbers and Decimal Frations

3- Number System , HCF & LCF

Rules of Divisibility

Divisibility rule of 2	Any number whose last digit is an even number (0, 2, 4, 6, 8) is divisible by 2
Divisibility rule of 3	A number is divisible by 3 if the sum of its digits is divisible by 3.
Divisibility rule of 4	A number is divisible by 4, if the number formed by the last two digits is divisible by 4.
Divisibility rule of 5	A number is exactly divisible by 5 if it has the digits 0 or 5 at one's place.
Divisibility rule of 6	A number is exactly divisible by 6 if that number is divisible by 2 and 3 both. It is because 2 and 3 are prime factors of 6.
Divisibility rule of 7	Double the last digit and subtract it from the remaining leading truncated number to check if the result is divisible by 7 until no further division is possible
Divisibility rule of 8	If the last three digits of a number are divisible by 8, then the number is completely divisible by 8.
Divisibility rule of 9	It is the same as of divisibility of 3. Sum of digits in the given number must be divisible by 9.
Divisibility rule of 11	If the difference of the sum of alternative digits of a number is divisible by 11, then that number is divisible by 11.
Divisibility rule of 12	A number is exactly divisible by 12 if that number is divisible by 3 and 4 both.
Divisibility rule of 13	Multiply the last digit with 4 and add it to remaining number in a given number, the result must be divisible by 13.
Divisibility rule of 15	If the number divisible by both 3 and 5, it is divisible by 15.
Divisibility rule of 17	Multiply the last digit with 5 and subtract it from remaining number in a given number, the result must be divisible by 17
Divisibility rule of 19	Multiply the last digit with 2 and add it to remaining number in a given number, the result must be divisible by 19.

Question 1 :Calculate how many numbers between 1 and 100, including both are divisible by 9 or 4.

A. 35
B. 33
C. 34
D. 30

ans: 34

Question 2 : Which of the following is exactly divisible by 11?

A. 817425
B. 4832817
C. 817259
D. 5533935

ans : 5533935

Question 3 : What will be the sum of remainders when 684 will be divided by 3 , 7 and 5?

A. 10
B. 9
C. 11
D. 6

ans: 9

Question 4 : Find the number from the given numbers which is not divisible by 7:

A. 84
B. 126
C. 89
D. 161

ans: 89

Question 5 : Find the second-smallest 3 digit number divisible by 2 and 7

A. 114
B. 128
C. 142
D. 107

ans: 128

Model Paper

Question 1- If a positive integer n is divided by 5, the remainder is 3. Which of the numbers below yields a remainder of 0 when it is divided by 5?

A. n+3

B. n+2

C. n-1

D. n-2

E. n+1

Question 2- **If an integer n is divisible by 3, 5 and 12, what is the next larger integer divisible by all these numbers?**

A. n+3

B. n+5

C. n+12

D. n+60

E. n+15

Question 3- **Which of there numbers is not divisible by 3?**

A. 339

B. 342

C. 552

D. 1111

E. 672

Question 4- **What is the smallest positive 2-digit whole number divisible by 3 and such that the sum of its digits is 9?**

A. 27

B. 33

C. 72

D. 18

E. 90

Question 5- **123456789101112.... . 434445 . Find the remainder when divided by 45?**

A. 5

B. 1

C. 3

D. Zero

Question 6- **If n is an integer, when $(2n + 2)^2$ is divided by 4 the remainder is**

A. zero

B. 2

C. 3

D. 4

Question 7- **Let N = 80pq2pq (7digit number). If N is exactly divisible by 120 then the sum of the digits in N is equal to:**

A. 18

B. 22

C. 24

D. 12

Question 8- **If 53p26p3 is a 7 digit number divisible by 9 and if 757qp is divisible by 8 then the minimum value of p + q is:**

A. 4

B. 9

C. 12

D. 16

Question 9- **Find the remainder when $1201 \times 1203 \times 1205 \times 1207$ is divided by 6.**

A. 5

B. 3

C. 9

D. 6

Question 10- **Find the remainder when $1! + 2! + 3! + 4! + 5! + ...100!$ is divided by 24.**

A. 5

B. 11

C. 7

D. 9

Explanation

Question -1

First, let's understand what it means when a number n divided by 5 has a remainder of 3:

It means n can be expressed as: $n = 5q + 3$, where q is some integer.

Now, we need to find a number that, when added to n, will make it divisible by 5 (i.e., have a remainder of 0).

Let's call this number x. So we're looking for:

$(n + x) \div 5$ should have a remainder of 0

Mathematically:

$(5q + 3 + x) \div 5$ should be a whole number

For this to be true, $(3 + x)$ must be divisible by 5

The smallest positive value of x that makes this true is 2, because:

$3 + 2 = 5$, which is divisible by 5

Therefore, adding 2 to n will always result in a number divisible by 5

Question -2

Let's approach this step-by-step:

First, we need to find the least common multiple (LCM) of 3, 5, and 12.

To find the LCM:

12 is already divisible by 3, so we only need to consider 12 and 5.

LCM(12, 5) = 12 * 5 = 60

This means that any number divisible by 3, 5, and 12 must be divisible by 60.

So, if n is divisible by 3, 5, and 12, we can express it as:

n = 60k, where k is some integer.

The next larger integer divisible by all these numbers would be:

n + 60

Therefore, the next larger integer divisible by 3, 5, and 12 after n is n + 60.

Question -3

339: 3 + 3 + 9 = 15
15 is divisible by 3, so 339 is divisible by 3.

342: 3 + 4 + 2 = 9
9 is divisible by 3, so 342 is divisible by 3.

552: 5 + 5 + 2 = 12
12 is divisible by 3, so 552 is divisible by 3.

1111: 1 + 1 + 1 + 1 = 4
4 is not divisible by 3, so 1111 is not divisible by 3.

672: 6 + 7 + 2 = 15
15 is divisible by 3, so 672 is divisible by 3.

Therefore, the number that is not divisible by 3 is 1111

Question -4

Let's approach this step-by-step:

First, we know we're looking for a 2-digit number, so it's between 10 and 99.

The number must be divisible by 3. A number is divisible by 3 if the sum of its digits is divisible by 3.

We're told that the sum of the digits is 9, which is indeed divisible by 3. So this condition is automatically satisfied.

Now, let's list out all the 2-digit numbers whose digits sum to 9:

18, 27, 36, 45, 54, 63, 72, 81, 90

Among these, we need to find the smallest one that's divisible by 3.

We can check each number:

$18 \div 3 = 6$ (no remainder)

We don't need to check further because 18 is the smallest in our list and it satisfies all conditions.

Therefore, the answer is 18. It's the smallest 2-digit number that's divisible by 3 and has digits that sum to 9.

Question -5

Let's approach this problem step by step:

First, we need to understand the pattern of the number:
123456789101112...434445
It's a concatenation of integers from 1 to 45.

To find the remainder when divided by 45, we can use the property that the sum of digits of a number divided by 9 gives the same remainder as the number itself divided by 9.

Let's find the sum of all digits:
1+2+3+4+5+6+7+8+9+1+0+1+1+1+2...+4+3+4+4+4+5

We can simplify this by using the formula for the sum of digits from 1 to n:
Sum of digits from 1 to n = n(n+1)/2

In this case, n = 45
Sum = 45(45+1)/2 = 45*46/2 = 1035

Now, we need to find the remainder when 1035 is divided by 45.
1035 ÷ 45 = 23 remainder 0

Therefore, the original number is divisible by 45, and the remainder is 0.

The remainder when the given number is divided by 45 is 0.

Question -6

Let's approach this step-by-step:

We start with the expression $(2n + 2)^2$.

Let's expand this:

$$(2n + 2)^2 = 4n^2 + 8n + 4$$

Now, we need to consider what happens when we divide this by 4. In division, we can separate terms:

$$(4n^2 + 8n + 4) \div 4 = n^2 + 2n + 1$$

The n^2 and $2n$ terms will always be divisible by 1 (as n is an integer), so they don't contribute to the remainder.

Therefore, the remainder when $(2n + 2)^2$ is divided by 4 will be the same as the remainder when 1 is divided by 4.

$1 \div 4 = 0$ remainder 1

Therefore, regardless of what integer n is, when $(2n + 2)^2$ is divided by 4, the remainder will always be 1.

Question -7

Let's approach this step-by-step:

First, we need to understand what the number N looks like:

N = 80pq2pq, where p and q are single digits.

So, N is a 7-digit number in the form 80pq2pq.

We're told that N is exactly divisible by 120. Let's consider the factors of 120:

$120 = 2^3 \times 3 \times 5$

For N to be divisible by 120, it must be divisible by 2^3 (8), 3, and 5.

For a number to be divisible by 8, its last three digits must be divisible by 8.

The last three digits of N are 2pq.

For a number to be divisible by 5, its last digit must be 0 or 5.

So, q must be 0 or 5.

For a number to be divisible by 3, the sum of its digits must be divisible by 3.

Let's sum the digits of N:

$8 + 0 + p + q + 2 + p + q = 10 + 2p + 2q$

This sum must be divisible by 3.

We know q is either 0 or 5. Let's consider both cases:

If $q = 0$:

$10 + 2p + 2(0) = 10 + 2p$ must be divisible by 3

This is only possible if $p = 4$ or 7

If $q = 5$:

$10 + 2p + 2(5) = 20 + 2p$ must be divisible by 3

This is only possible if $p = 3$ or 6

Now, let's check which of these possibilities makes 2pq divisible by 8:

240 is divisible by 8

270 is not divisible by 8

235 is not divisible by 8

265 is divisible by 8

Therefore, there are two possibilities for N:

8040240 or 8065265

The sum of digits in both cases is:

$8 + 0 + 4 + 0 + 2 + 4 + 0 = 18$

$8 + 0 + 6 + 5 + 2 + 6 + 5 = 32$

Therefore, the sum of the digits in N is either 18 or 32.

Question -8

For the first condition: 53p26p3 is divisible by 9

A number is divisible by 9 if the sum of its digits is divisible by 9

Sum of digits = $5 + 3 + p + 2 + 6 + p + 3 = 19 + 2p$

For this to be divisible by 9, we need: $19 + 2p = 9k$, where k is some integer

$2p = 9k - 19$

$p = (9k - 19) / 2$

The smallest positive integer value for p that satisfies this is when $k = 3$

$p = (27 - 19) / 2 = 4$

For the second condition: 757qp is divisible by 8

For a number to be divisible by 8, its last three digits must form a number divisible by 8

In this case, the last three digits are 7qp

We know $p = 4$ from the first condition

So, we need to find q such that 7q4 is divisible by 8

Testing values:

- If $q = 0$, 704 is not divisible by 8
- If $q = 1$, 714 is not divisible by 8
- If $q = 2$, 724 is divisible by 8

Therefore, the minimum values are:

$p = 4$

$q = 2$

The minimum value of $p + q = 4 + 2 = 6$

Thus, the minimum value of $p + q$ is 6.

Question -9

Let's approach this step-by-step:

To find the remainder when a number is divided by 6, we can use the following properties:

If a number is divisible by both 2 and 3, it's divisible by 6.

The remainder when divided by 6 can be determined by considering the remainders when divided by 2 and 3 separately.

Let's look at each number modulo 2 and 3:

- $1201 \equiv 1 \pmod{2}$ and $1201 \equiv 2 \pmod{3}$
- $1203 \equiv 1 \pmod{2}$ and $1203 \equiv 0 \pmod{3}$
- $1205 \equiv 1 \pmod{2}$ and $1205 \equiv 2 \pmod{3}$
- $1207 \equiv 1 \pmod{2}$ and $1207 \equiv 1 \pmod{3}$

When we multiply these numbers:

- Modulo 2: $1 \times 1 \times 1 \times 1 \equiv 1 \pmod{2}$
- Modulo 3: $2 \times 0 \times 2 \times 1 \equiv 0 \pmod{3}$

So, our result is odd (remainder 1 when divided by 2) and divisible by 3.

When a number leaves remainder 1 when divided by 2 and remainder 0 when divided by 3, it leaves remainder 3 when divided by 6.

Therefore, the remainder when $1201 \times 1203 \times 1205 \times 1207$ is divided by 6 is 3.

Question -10

Let's approach this step-by-step:

First, let's look at the pattern of factorials modulo 24:

$1! = 1$
$2! = 2$
$3! = 6$
$4! = 24 \equiv 0 \pmod{24}$
$5! = 120 \equiv 0 \pmod{24}$
$6! = 720 \equiv 0 \pmod{24}$
...and so on.

We can see that from 4! onwards, all factorials are divisible by 24.

This means in our sum, all terms from 4! to 100! will contribute 0 to the remainder.

So, we only need to consider $1! + 2! + 3!$

Let's calculate this:

$1! + 2! + 3! = 1 + 2 + 6 = 9$

Therefore, the remainder when $1! + 2! + 3! + 4! + 5! + ... + 100!$ is divided by 24 is the same as the remainder when 9 is divided by 24.

The remainder when 9 is divided by 24 is 9.

Therefore, the final answer is 9